Demonstration of Proposed Features

Project Information

Date	15 March 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features

This document captures all the features that were proposed during the project planning phase for **OptiCrop: Smart Agricultural Production Optimization Engine** and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver a completed machine learning based agricultural decision-support solution.

S.No	Feature Name	Description	Status	Demonstrated	Remarks
1	Crop Recommendation	Predicts the most suitable crop using soil nutrients and weather parameters.	Implemented	Yes	Recommendation result displays successfully with crop name.
2	Confidence Score	Shows the prediction confidence generated by the trained machine learning model.	Implemented	Yes	Confidence value is displayed along with the recommended crop.
3	Input Validation	Validates Nitrogen, Phosphorus, Potassium, temperature, humidity, pH, and rainfall values.	Implemented	Yes	Prevents invalid or incomplete user input.

4	Prediction History	Stores previous crop prediction records in the SQLite database.	Implemented	Yes	History records persist and can be reviewed from the dashboard.
5	Dashboard	Displays prediction results, previous records, and basic agricultural insights.	Implemented	Yes	Dashboard presents results in a clear and user-friendly format.
6	Responsive Web Interface	Provides a Bootstrap-based interface for desktop and mobile users.	Implemented	Yes	Application layout adjusts properly across screen sizes.

Feature Implementation Summary

Metric	**Value**
Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%